Multiple dimensions of network structure underlie ecological stability

Tanya Strydom [1]; Andrew P. Beckerman [1]

Abstract: Ecological network structure is widely invoked to explain patterns of biodiversity and stability, yet decades of research have produced little consensus on which structural descriptors matter most. We argue that this apparent inconsistency arises not from methodological disagreement, but from a conceptual mismatch: network structure encodes multiple, hierarchically organised aspects of energy flow, each relating to different components of stability. Using a compilation of food webs from XXX, we quantified 31 commonly used structural metrics spanning species roles, interaction pathways, network geometry, and emergent system behaviour. We first used variable clustering and dimensionality reduction to identify dominant axes of structural variation and a non-redundant subset of descriptors. We then interpreted these axes through an energy-flow framework linking structural scale to distinct stability mechanisms: persistence (maintenance of energy flow following species loss), resistance (attenuation of perturbation propagation), and return (reorganisation of pathways following disturbance). Our analyses show that metrics often treated as competing predictors of 'stability' instead load onto different structural dimensions and relate to different stability components. Redundancy and species-role metrics primarily align with persistence, pathway and motif-based metrics with resistance, and global organisational measures with return dynamics. By explicitly mapping network structure to stability mechanisms, our results reconcile conflicting findings in the structure–stability literature and provide a principled framework for selecting network metrics based on ecological question and scale, rather than convention.

Keywords: food web, structure, dimensionality reduction

1 Introduction

Which metrics best represent the distinct stages of the energy-flow hierarchy, and do they capture different components of stability?

Food web structure self-organises into process-level modules, and those modules govern different components of stability.

Food web architecture is organised into emergent structural process modules, and different modules regulate different components of ecological stability.

Food web metrics self-organise into structural process modules, these modules define network space, and different modules regulate different stability components.

Should we be looking at if different metrics scale predictably with each other?

Ecological networks are commonly characterised using a large and diverse set of structural metrics, yet there remains little consensus on how these metrics should be interpreted or compared across studies (Vermaat et al. 2009; Lau et al. 2017). We argue that this difficulty arises because network structure is not a single property, but a hierarchy of interrelated components that encode different aspects of how energy moves through an ecosystem (Ulanowicz 1986; Thompson et al. 2012). Metrics derived at different structural scales therefore capture different ecological processes and, consequently, relate to different components of stability. Making these distinctions explicit is essential for interpreting patterns of structure–stability relationships.

1.0.1 Structural scale and the organisation of energy flow

We conceptualise food-web structure as organised across four hierarchical structural scales: node-level, path-level, network geometry, and emergent system behaviour (Pimm 1982; Cohen et al. 1990; Vermaat et al. 2009). These scales reflect increasing levels of integration, progressing from individual species roles to whole-network dynamics.

Node-level structure: describes the properties and roles of individual species within the network. Metrics such as trophic position, basal and top species proportions, centrality, and trophic similarity capture how energy enters the system, which species process or redistribute that energy, and the degree of functional redundancy among taxa (Yodzis & Winemiller 1999; Estrada 2007; Allesina & Pascual 2009). At this scale, structure primarily reflects who participates in energy transfer and whether alternative species can compensate for local losses, a long-standing mechanism proposed to underpin persistence in complex ecosystems (McCann 2000).

Path-level structure: describes how energy is routed through sequences of interactions linking species. Metrics such as food chain length, omnivory, motifs, loops, and prey–predator ratios capture the coupling of energy channels and the multiplicity of pathways connecting basal resources to higher trophic levels (Pimm & Lawton 1977; McCann et al. 1998; Stouffer & Bascompte 2011). This scale reflects how energy moves through the network and how perturbations may be transmitted across species, with particular pathway configurations either amplifying or diffusing disturbance effects (Neutel et al. 2002; Rooney et al. 2006).

Network geometry: captures the global arrangement and constraints of interactions, including link density, connectance, clustering, modularity, intervality, and network distances (Dunne et al. 2002; Stouffer et al. 2006; Delmas et al. 2019). These properties define the overall "shape" of the food web and constrain which pathways are available for energy flow. Geometric organisation has been shown to influence the containment of perturbations, for example through compartmentalisation that limits the spread of local disturbances (Stouffer & Bascompte, 2011).

Emergent system-level behaviour: reflects the collective dynamical properties that arise from the interaction of nodes, paths, and geometry. Metrics such as spectral radius, SVD complexity, and robustness capture properties that cannot be attributed to individual species or interactions alone, but instead describe the system's overall capacity to absorb, reorganise, or amplify perturbations (May 1972; Staniczenko et al. 2013; Strydom et al. 2021).

Together, these structural scales form a causal hierarchy: species roles give rise to interaction pathways, which are embedded within a global network geometry, from which emergent dynamical behaviour arises.

1.0.2 Stability as a multi-component property

Stability is often treated as a single outcome, yet ecological theory has long recognised that it comprises multiple, distinct components (Pimm 1984; Ives & Carpenter 2007). Here, we focus on three complementary stability mechanisms that are directly interpretable in terms of energy flow: persistence, resistance, and return.

Persistence: refers to the continued ability of the system to sustain energy flow following species loss or disturbance. Structures that promote redundancy, multiple basal inputs, and functional overlap among species enhance persistence by ensuring that energy can continue to enter and circulate within the network even when individual components are removed (McCann 2000; Dunne et al. 2002; Jonsson et al. 2015).

Resistance: describes the extent to which perturbations are attenuated or contained rather than propagating through the network. Path-level structures such as omnivory, short chains, motifs, and modular organisation

can diffuse or localise disturbances, reducing the likelihood that local perturbations escalate into system-wide effects (McCann et al. 1998; Neutel et al. 2002; Stouffer & Bascompte 2011).

Return: captures the capacity of the system to reorganise or recover following disturbance, including the re-establishment of interaction pathways and the damping of oscillations. Emergent properties linked to global organisation and heterogeneity, such as spectral radius and structural complexity, reflect constraints on system-wide dynamics and influence the speed and manner with which a system returns to equilibrium or a new stable configuration (May 1972; Ulanowicz 2001; Staniczenko et al. 2013).

Crucially, these stability components operate at different structural scales and are not expected to respond uniformly to the same network properties.

1.0.3 Implications for interpreting network metrics

Within this framework, metrics that are often treated as competing predictors of 'stability' instead emerge as complementary descriptors of different stability mechanisms (Thompson et al., 2012). Node-level metrics primarily relate to persistence, path-level metrics to resistance, and global organisational metrics to return dynamics. Some descriptors span multiple scales, reflecting the coupling between structural organisation and emergent behaviour (Allesina & Tang 2012).

This perspective provides a mechanistic explanation for why studies using different network metrics frequently report contrasting structure–stability relationships. Rather than reflecting inconsistency or redundancy, these differences arise because different metrics implicitly target different components of stability (Lau et al., 2017). By explicitly linking structural scale, energy flow, and stability mechanism, this framework provides a principled basis for interpreting network metrics and for selecting descriptors that align with specific ecological questions.

Table 1: Words

Structural scale	What it encodes	Energy-flow interpretation	Stability component
Node	Species roles and redundancy	Who handles energy	Persistence (can energy still enter and move?)
Path	Energy routing and coupling	How energy moves	Resistance (does disturbance spread?)
Geometry	Network constraints and organisation	Where energy can go	Containment / buffering

Structural scale	What it encodes	Energy-flow interpretation	Stability component
Behaviour	Emergent dynamics	How energy reorganizes	Return / reassembly

2 Materials & Methods

2.1 Data Compilation

We compiled quantitative network data from XX, resulting in a total of XX ecological networks. Each network was characterized using a suite of XX structural metrics Table 2, including classic descriptors such as richness, connectance, and modularity, as well as information-theoretic measures like spectral radius and SVD complexity. Prior to analysis, networks with missing values were omitted, and all metrics were standardized (mean = 0, SD = 1) to account for differences in scale and units across descriptors.

Table 2: An informative caption about the different network properties. We use a combination of metrics from both the original Vermaat et al. (2009) paper as well as including those that have been identified by Thompson et al. (2012) and have been linked to emerging ecosystem properties such as stability

Label	Definition	Structural interpretation	Reference
Basal	Proportion of taxa with zero vulnerability (no consumers).	Quantifies the proportion of species representing basal energy inputs to the network.	
Top	Proportion of taxa with zero generality (no resources).	Describes the relative prevalence of terminal consumers in the network.	
Intermediate	Proportion of taxa with both consumers and resources.	Captures the proportion of species participating in both upward and downward energy transfer.	
Richness (S)	Number of taxa (nodes) in the network.	Describes network size.	
Links (L)	Total number of trophic interactions (edges).	Describes interaction density independent of network size.	

Label	Definition	Structural interpretation	Reference
Connectance	L/S^2, where S is the number of species and L the number of links	Measures the proportion of realised interactions relative to all possible interactions.	Dunne et al. 2002
L/S	Mean number of links per species.	Captures average interaction density per taxon.	
Cannibal	Proportion of taxa with self-loops.	Quantifies the prevalence of cannibalistic interactions.	
Herbivore	Proportion of taxa feeding exclusively on basal species.	Describes the representation of primary consumers.	
Intermediate	Percentage of intermediate taxa (with both consumers and resources)		
Trophic level (TL)	Prey-weighted trophic level averaged across taxa.	Captures the vertical organisation of energy transfer.	Williams & Martinez (2004)
MaxSim	Mean maximum trophic similarity of each taxon to all others.	Quantifies functional similarity based on shared predators and prey.	Yodzis & Winemiller (1999)
Centrality	Node centrality averaged across taxa (definition-dependent).	Captures the distribution of influence or connectivity among species.	Estrada & Bodin (2008)
ChLen	Mean length of all food chains from basal to top taxa.	Describes the average number of steps in energy-transfer pathways.	
ChSD	Standard deviation of food chain length.	Captures variability in pathway lengths.	

Label	Definition	Structural interpretation	Reference
ChNum	Log-transformed number of distinct food chains.	Quantifies the multiplicity of alternative energy pathways.	
Path	Mean shortest path length between all species pairs.	Describes the average distance between taxa within the network.	
Diameter	Maximum shortest path length between any two taxa.	Captures the largest network distance between species.	
Omnivory	Proportion of taxa feeding on resources at multiple trophic levels.	Describes vertical coupling of energy channels.	McCann (2000)
Loop	Proportion of taxa involved in trophic loops.	Quantifies the prevalence of cyclic interaction pathways.	
Prey:Predator	Ratio of prey taxa (basal + intermediate) to predator taxa (intermediate + top).	Describes the overall shape of the trophic structure.	
Diameter	Diameter can also be measured as the average of the distances between each pair of nodes in the network		
Clust	Mean clustering coefficient.	Measures the tendency for taxa sharing interaction partners to also interact with each other.	Watts & Strogatz (1998)
GenSD	Normalised standard deviation of generality.	Captures heterogeneity in the number of resources per taxon.	Williams & Martinez (2004)

Label	Definition	Structural interpretation	Reference
VulSD	Normalised standard deviation of vulnerability.	Captures heterogeneity in the number of consumers per taxon.	Williams & Martinez (2004)
LinkSD	Normalised standard deviation of total links per taxon.	Quantifies variation in species connectivity.	
Intervality	Degree to which taxa can be ordered along a single niche dimension.	Measures the extent of niche ordering in trophic interactions.	Stouffer et al. (2006)
(Spectral radius)	Largest real part of the eigenvalues of the undirected adjacency matrix.	Captures a global property of network organisation related to interaction strength aggregation.	
Complexity (SVD)	Shannon entropy of the singular value decomposition of the adjacency matrix.	Quantifies heterogeneity in interaction structure.	Strydom et al. (2021)
Robustness	Proportion of secondary extinctions following primary species removal.	Operational measure of tolerance to node loss.	Jonsson et al. (2015)
S1 (Linear chain)	Frequency of three-node linear chains (A $\rightarrow$ B $\rightarrow$ C) with no additional links.	Captures the prevalence of simple, unbranched energy-transfer pathways.	Stouffer et al. (2007) Milo et al. (2002)
S2 (Omnivory)	Frequency of three-node motifs forming a feed-forward loop (A $\rightarrow$ B $\rightarrow$ C, A $\rightarrow$ C).	Describes vertical coupling of trophic levels within small subnetworks.	Stouffer et al. (2007) Milo et al. (2002)

Label	Definition	Structural interpretation	Reference
S4 (Apparent competition)	Frequency of motifs where one consumer feeds on two resources (A → B ← C).	Captures the prevalence of shared-predator structures among resources.	Stouffer et al. (2007) Milo et al. (2002)
S5 (Direct competition)	Frequency of motifs where two consumers share a single resource (A ← B → C).	Describes the occurrence of shared-resource structures among consumers.	Stouffer et al. (2007) Milo et al. (2002)

2.2 Identification of Structural Modules

We tested the hypothesis that structural descriptors of food webs are organised into statistically coherent modules reflecting shared ecological function or scaling relationships, rather than forming arbitrary clusters driven by sampling noise. If such modular organization exists, then metrics within a module should exhibit strong internal correlation relative to metrics assigned to different modules. The resulting clusters should be robust to resampling of the data. The identified modular structure should exceed expectations under null models of random association among metrics.

We quantified pairwise associations among the 30 structural metrics using Pearson correlations computed across food webs. Because ecological metrics may covary either positively or negatively depending on scaling relationships, we constructed two alternative distance matrices: $1 - r$, which preserves the sign of correlations and distinguishes positive from negative association and $1- \mid r \mid$, which groups variables based on the magnitude of their association regardless of sign. These two definitions allow us to test whether modular structure depends on directional relationships or simply on the strength of coupling among metrics. Hierarchical clustering was performed using average linkage on each distance matrix to identify candidate structural modules.

The optimal number of clusters was evaluated across a range of partition sizes (k = 2–10) using average silhouette width to assess within-cluster cohesion and between-cluster separation. To further evaluate cluster robustness, we implemented bootstrap resampling with 1,000 bootstrap replicates. This procedure estimates approximately unbiased (AU) p-values for each cluster, quantifying the probability that a cluster is supported under repeated resampling of the data. Clusters were considered statistically robust when they exhibited high silhouette support, and approximately unbiased bootstrap support 0.95. This dual criterion ensures that

identified modules are both structurally coherent and stable to sampling variation.

2.3 Multivariate Structure and Module–Axis Alignment

To evaluate whether structural metrics of food webs organize into coherent multivariate modules and whether those modules define principal axes of variation, we conducted a principal component analysis (PCA) followed by a permutation-based test of module–axis alignment. Under this framework, principal components define dominant structural gradients in the metric space, whereas modules represent hypothesised mechanistic groupings of structurally related metrics. Demonstrating alignment between these two representations would suggest that modular decomposition captures fundamental axes of ecological variation.

Skewed count-based metrics (*e.g.,* link counts, interval counts, and related quantities) were log-transformed using $log(x + 1)$ to reduce right skew. All remaining variables were standardized to zero mean and unit variance prior to analysis to ensure that metrics measured on different scales contributed equally to the ordination.

To quantify how structural modules align with principal axes, we decomposed variance in each principal component according to module membership. For each module m and principal component k, we computed:

$$A_{mk} = \sum_{i \in m} l_{ik}^2$$

where l_{ik} is the loading of metric i on principal component k and A_{mk} represents the fraction of variance in PC k attributable to metrics in module m. Because squared loadings sum to one within each principal component, this provides a direct partitioning of PC variance across modules. This produces a module $\times$ PC matrix describing geometric alignment between modular structure and multivariate axes.

To evaluate whether observed module–PC alignment exceeded expectations under random module structure, we implemented a permutation test. We randomly permuted module labels among metrics 1 000 times while preserving the number and size distribution of modules, and the PCA loadings. For each permutation r, we recomputed $A_{mk}^{(r)}$ so as to form a null distribution for each module–PC pair. From this we computed p-values and corresponding z-scores, which was then used to infer significant alignment when $p_{mk} < 0.05 \wedge |Z_{mk}| > 1.96$

$$p_{mk} = P(A_{mk}^{(r)} > A_{mk})$$

$$Z_{mk} = \frac{A_{mk} - \mu_{mk}}{\sigma_{mk}}$$

Additionally we also evaluated if modules collectively aligned with PCA structure beyond random expectation. Overall concentration of variance within module–PC space was calculated as $T = \sum_{m,k} A_{mk}^2$ and significance was determined by comparing the observed (T) the permutation distribution ($T^{(r)}$). Where $p_{global} = P(T^{(r)} \geq T)$. A significant result indicates that module structure explains multivariate variance better than random groupings.

2.4 Structural Metric Reduction and Representation of Network Space

To evaluate how network structure predicts ecological stability, we first reduced a high-dimensional set of topological metrics into alternative, biologically interpretable predictor sets. Because dimensionality reduction can fundamentally shape inference, we explicitly compared two conceptually distinct representations of network structure: (1) structural domain representatives and (2) latent axes of network variation.

2.4.1 Identification of Structural Domains

We previously grouped network metrics into clusters based on their pairwise correlations, such that each cluster represented a structurally coherent domain (e.g., trophic composition, centralization, path structure). Clustering was performed using correlation-based distance, ensuring that metrics grouped together reflected shared structural information rather than raw scale. To construct a reduced predictor set from these domains, we selected a single representative metric per cluster using a medoid approach. Within each cluster, we computed the absolute correlation matrix among member metrics and defined distance as 1− r . The medoid was identified as the metric minimizing mean pairwise distance to other metrics in the cluster. This approach preserves structural diversity while minimizing redundancy and does not rely on principal component loadings, which can bias selection toward dominant variance axes. The resulting 'cluster medoid' set retained one interpretable metric per structural domain. Importantly, this procedure prioritizes preservation of structural domains rather than overall variance magnitude, allowing low-variance but potentially mechanistically relevant features of network topology to be retained.

2.4.2 Extraction of Latent Structural Axes

As a complementary representation of network structure, we performed principal component analysis (PCA) on the scaled full metric matrix. Principal components were retained until cumulative explained variance exceeded 80%, yielding a set of orthogonal axes describing the dominant gradients of variation in network topology.

These retained PC scores constitute a low-dimensional latent representation of network space. Unlike cluster medoids, which preserve structural categories, PC scores preserve maximal variance and ensure orthogonality among predictors. We also explored a third reduction approach—selecting the single highest-loading metric per retained PC. However, because principal components are linear combinations of multiple metrics, this "PC-dominant metric" strategy substantially reduces the variance represented by each axis. Consequently, this approach captures less of the total structural variance and was treated as exploratory rather than primary.

2.4.3 Conceptual Contrast Between Representations

The cluster-medoid and PCA-score representations reflect different theoretical assumptions about how structure relates to stability. The cluster-medoid approach assumes that ecological stability responds to discrete structural domains that may vary independently and need not align with the dominant gradients of network variation. In contrast, the PCA-score approach assumes that stability responds primarily to the major axes of variation in network topology, regardless of their interpretability in terms of structural domains. By explicitly comparing predictive performance across these representations, we test whether ecological stability is better explained by domain-specific structural mechanisms or by global geometric gradients of network organization.

2.4.4 Implications for Model Comparison

Because the cluster-medoid and PCA-score sets differ in dimensionality and variance preservation, predictive comparisons were conducted using identical cross-validation schemes to ensure fair evaluation. Model performance was assessed using out-of-sample metrics, and variable importance was evaluated via cross-validated permutation procedures. This framework allows us to determine whether stability aligns more strongly with interpretable structural domains or with latent axes of network variation, and whether some structural domains are decoupled from stability altogether.

3 Results

3.1 Structural Metrics Organise into Robust Modules

Hierarchical clustering of structural descriptors revealed clear modular organisation within food-web architecture Figure 1. Silhouette analysis indicated an optimal partition of the signed correlation matrix at k=7 modules, with a maximum average silhouette width of [insert value]. Clustering based on absolute correlations produced a slightly coarser but comparable solution (k=5), demonstrating that the modular structure is

robust to the treatment of correlation sign and reflects strong underlying association patterns among metrics.

[Figure 1 about here.]

Bootstrap resampling further supported the stability of the major clusters. Most principal modules exhibited high approximately unbiased (AU) support values (AU [insert value]), indicating that the identified clusters are not artefacts of sampling variability but represent statistically robust groupings of structural descriptors.

The resulting modular partition identified coherent groups of metrics describing distinct aspects of network architecture. These included a large-scale architectural module integrating size and motif-based descriptors, modules capturing trophic organisation and energy routing, and two modules consisting of isolated structural descriptors that function independently from broader metric groupings. Importantly, the modular structure persisted after accounting for species richness, demonstrating that clustering was not driven solely by network size effects.

3.2 Modular Decomposition of Structural Space

The seven-module solution revealed a hierarchical decomposition of structural space into interpretable architectural dimensions. Collectively, the module decomposition reveals that food web structure can be interpreted as statistically supported and ecologically interpretable dimensions of network organisation that is strongly suggestive of a delineation by hierarchical architecture.

Macro-Architectural Complexity: The largest module encompassed species richness, number of interactions, diameter, motif frequencies, intervality, and related descriptors. These metrics jointly describe large-scale expansion of the interaction network and increased combinatorial complexity as species richness increases. This module reflects macro-architectural scaling: increasing network size simultaneously expands motif diversity, path structure, and global trophic ordering.

Trophic Integration: A second module grouped metrics describing vertical trophic organisation and connectivity, including connectance, trophic level, omnivory, chain length, and clustering. Together, these descriptors quantify how densely and hierarchically energy and interactions are embedded across trophic strata. This module captures the degree to which the network integrates trophic levels through cross-level interactions and structural embedding.

Energy Transport: A distinct module isolated geometric descriptors of interaction pathways, including mean distance, path number, and dispersion in chain length. These metrics characterize the structural geometry of energy routing independent of overall network size. The separation of this module suggests that routing structure constitutes an independent dimension of architectural variation rather than merely scaling with

complexity.

Trophic Asymmetry: Another module captured metrics describing structural imbalance between basal and consumer components, including the proportion of basal species, predator–prey ratios, and generality measures. This module reflects bottom-heavy versus top-heavy structural skew and quantifies asymmetry in the distribution of trophic roles across the network.

Control Heterogeneity: A separate module grouped descriptors of heterogeneity in consumer pressure and top-down control, including the proportion of apex species, herbivory structure, and variance in vulnerability. This module captures variation in how evenly or unevenly control is distributed across trophic levels.

Centralisation and Functional Redundancy: Two metrics formed independent modules. Centrality emerged as a standalone module, representing concentration of structural dominance within a subset of species. Maximum trophic similarity likewise formed an isolated module, quantifying functional redundancy and niche overlap. The separation of these descriptors from broader structural groupings indicates that dominance concentration and trophic redundancy represent independent axes of food-web organisation.

Collectively, these seven modules delineate a hierarchical decomposition of food-web structure spanning macro-scale complexity, vertical integration, routing geometry, trophic imbalance, control heterogeneity, dominance concentration, and functional overlap.

Table 3: Structural Modules Identified From Hierarchical Clustering.

Module Name	Metrics Included	Ecological Interpretation	Robustness (Silhouette / AU)	Module Size
Macro-Architectural Complexity	richness, links, diameter, motif frequencies, intervality, path-related metrics	Captures large-scale expansion of network structure and combinatorial complexity associated with increasing system size		
Trophic Integration	connectance, trophic level, omnivory, chain length, clustering	Describes vertical trophic embedding and density of cross-level interactions		

Module Name	Metrics Included	Ecological Interpretation	Robustness (Silhouette / AU)	Module Size
Energy Transport	mean distance, path number, chain-length dispersion	Quantifies geometric structure of energy routing independent of size scaling		
Trophic Asymmetry	basal proportion, predator–prey ratio, generality	Represents structural skew between basal and consumer dominance; bottom-heavy vs top-heavy organization		
Control Heterogeneity	apex proportion, herbivory, vulnerability variance	Captures heterogeneity in top-down control and uneven distribution of interaction pressure		
Centralisation	centrality	Measures concentration of structural dominance within a subset of species		
Functional Redundancy	maximum trophic similarity	Quantifies trophic overlap and niche redundancy; independent axis of functional equivalence		

3.3 Structural Modules Align with Dominant Axes of Network Variation

Principal component analysis of standardised structural metrics revealed strong dimensional compression in food-web structure Figure 2. The first three principal components accounted for 69.1% of total variance (PC1 = 31.2%, PC2 = 22.4%, PC3 = 15.4%), with 80.8% of variance explained by the first five components. Subsequent axes contributed progressively smaller proportions of variance, indicating that a small number of dominant gradients summarise most structural variation.

[Figure 2 about here.]

Inspection of the loading structure showed that the first principal component primarily reflected variation in

network size and connectivity. Metrics such as richness, number of links, diameter, path length, and measures associated with topological complexity loaded strongly on this axis. The second principal component captured variation associated with structural organization and connectivity patterns, whereas the third component reflected variation in trophic structure and vertical differentiation, with strong contributions from trophic level, chain length, and related measures. Together, these axes represent interpretable structural gradients in the metric space.

[Figure 3 about here.]

The global permutation test indicated significant alignment between the modular partition and the principal component structure ($P = 0.005$), demonstrating that the observed module assignment explains the multivariate geometry of structural metrics better than randomly assigned modules. Alignment strength varied substantially across modules Figure 4. Macro Complexity (Module 1) exhibited the strongest and most extensive alignment, showing significant associations with five principal components and a dominant loading on PC1, indicating that it contributes substantially to the primary structural gradient associated with network size and connectivity. Energy Transport (Module 3) showed significant alignment with four principal components, with its strongest association on PC4, suggesting that it captures structural variation largely orthogonal to the dominant complexity axis. Trophic Integration (Module 2), Trophic Asymmetry (Module 4), Control Heterogeneity (Module 5), and Functional Redundancy (Module 7) displayed more limited alignment, each significantly associated with one to two principal components, indicating more constrained mapping onto the structural feature space. Notably, Functional Redundancy exhibited its strongest association with a high-order principal component (PC7), suggesting sensitivity to more subtle structural variation. Centralisation (Module 6) showed no significant alignment under the joint permutation and Z-score criterion, indicating that it does not meaningfully structure variation across the retained principal components. Overall, significant module–PC associations were concentrated in a subset of principal components, reinforcing the interpretation that modular structure maps onto specific structural gradients rather than uniformly spanning the full multivariate space.

[Figure 4 about here.]

Collectively, these results demonstrate that food-web structural metrics exhibit strong low-dimensional organisation and that independently identified structural modules are non-randomly aligned with the dominant axes of variation. The significant global alignment supports the interpretation that modular decomposition reflects meaningful geometric structure rather than clustering artifacts. Moreover, the dominance of a single module across multiple principal components suggests hierarchical organisation, wherein a core structural module governs the primary gradient of variation while additional modules capture secondary and specialised

structural dimensions.

3.4 Predictors of stability

3.4.1 Dimensional Reduction Characterisation

Representation	Dimensionality	Variance Preserved	Structure Type
Medoids	7	23%	Domain sampling
PC-dominant metrics	4	13%	Axis proxy sampling
PC scores	5	80%	True latent space

To evaluate how alternative structural representations differed in their information content and redundancy, we quantified variance retention, internal correlation structure, effective dimensionality, and geometric similarity among representations.

Variance Retention: The three representations differed substantially in the proportion of total structural variance preserved (Fig. S1A). The cluster-medoid representation retained ~23% of the total variance in the full metric space despite reducing dimensionality to seven predictors. In contrast, the PC-dominant metric set retained only ~13% of total variance, reflecting the fact that individual metrics capture only a fraction of each principal component's multivariate structure. As expected by construction, the retained PC-score representation preserved approximately 80% of total variance. Thus, domain-based reduction preserved more distributed structural information than selecting one metric per PC axis, whereas the PC-score representation maximally preserved dominant gradients of network variation.

Internal Redundancy: Representations also differed in their internal correlation structure (Fig. S1B). PC scores were orthogonal by definition (mean |r| 0), indicating complete statistical independence among predictors. In contrast, cluster medoids exhibited moderate residual correlation (mean |r| = X), suggesting partial overlap among structural domains. PC-dominant metrics showed comparable (or higher/lower — insert result) redundancy relative to medoids. These differences indicate that the three approaches vary not only in information retention but also in predictor independence.

Effective Dimensionality: We next quantified effective dimensionality as the number of axes required to explain 80% of variance within each reduced predictor set (Fig. S1C). The PC-score representation required X axes, reflecting its design to capture dominant structural gradients. The cluster-medoid representation required X axes to reach the same threshold, indicating that despite containing seven predictors, structural variation was concentrated along fewer effective dimensions. The PC-dominant set exhibited the lowest

effective dimensionality (X axes), consistent with its reduced variance retention. Together, these results show that dimensional compression differed across approaches not only in magnitude but in the distribution of variance across axes.

Overall, the three structural representations differed substantially in variance retention, redundancy, and effective dimensionality. Cluster medoids preserved moderate variance while maintaining domain interpretability, PC-dominant metrics retained minimal total variance, and PC scores preserved dominant structural gradients while ensuring predictor orthogonality. These differences establish that the representations encode distinct aspects of network topology, justifying their empirical comparison in predictive analyses of stability.

[Figure 5 about here.]

3.4.2 Predictive Performance Across Structural Representations

Structural Predictability of Stability Components: Network structural metrics predicted stability components with varying strength (Table 1). Predictive performance was highest for complexity, which achieved a cross-validated R^2 of 0.78 ± 0.14 (mean ± SD) under the cluster-medoid representation. Predictability was intermediate for (0.58 ± 0.20 under cluster medoids) and consistently low for robustness across all representations (maximum R^2 =0.25). Across stability components, robustness exhibited the lowest explanatory power (mean R^2 =0.14–0.25), whereas complexity showed the strongest and most consistent signal. These results indicate that not all dimensions of stability are equally encoded in network structure.

Comparison Among Structural Representations: Predictive performance differed among dimensional-reduction approaches, but effects were stability-component specific. For complexity, the cluster-medoid representation substantially outperformed PC-based alternatives. Cluster medoids achieved a mean R^2 =0.776 ± 0.139, compared to 0.543 ± 0.104 for PC-dominant metrics and 0.403 ± 0.251 for PC scores. A paired comparison between cluster medoids and PC scores revealed a significant performance difference (paired t-test: t_4 =5.01, p = 0.007), with a mean R^2 increase of 0.37. Thus, complexity was best predicted by domain-based structural descriptors rather than latent axes. In contrast, robustness showed low and statistically indistinguishable performance across representations (cluster: 0.189 ± 0.258; PC scores: 0.247 ± 0.240; paired t_4 =−0.35, p = 0.74). Structural representation choice had little influence on predictive ability for this component. For , cluster medoids again produced the highest mean performance (0.584 ± 0.204), followed by PC scores (0.404 ± 0.246) and PC-dominant metrics (0.384 ± 0.094). However, the difference between cluster medoids and PC scores was not statistically significant (paired t_4 = 1.36, p = 0.24). Taken together, these results indicate that structural domain representatives can outperform latent-axis representations, but this advantage is not universal across stability components.

Variability Across Folds: Fold-level variability also differed among representations. PC-score models exhibited higher variance in predictive performance (SD up to 0.25) relative to PC-dominant metrics (SD 0.11 across components), indicating less stable out-of-sample generalization. Cluster-medoid models showed intermediate variability, except for robustness, where fold variance was high across all approaches. These differences suggest that dimensional reduction strategy influences not only mean predictive power but also model stability.

[Figure 6 about here.]

3.4.3 Predictor Importance and Structural Drivers

3.4.3.1 Structural Drivers of Stability

Permutation-based variable importance revealed no single structural metric that exhibited statistically robust effects across stability components or structural representations (all $Z < 1$; Table Sx). Thus, despite differences in predictive performance, stability components were not attributable to singular structural drivers. For complexity, predictive signal was moderately distributed across metrics. Under the cluster representation, linkage density (l_S) contributed 47.8% of total importance, followed by maximum similarity (MaxSim), but neither exceeded permutation expectations. PC-based representations showed stronger apparent concentration (PC_Dominant: 63% attributed to S1; PC_Scores: 41% to PC1), yet these effects remained statistically weak. Robustness exhibited low overall predictive performance and correspondingly unstable importance structure. Although basal species fraction dominated the cluster model (57% contribution) and predator–prey ratio dominated the PC_Dominant model (>100% relative contribution due to negative importances among other predictors), no metric demonstrated consistent significance, indicating that apparent dominance reflects model instability rather than mechanistic control. For , importance was comparatively diffuse across predictors. Under the cluster representation, vulnerability heterogeneity (VulSD) and linkage density shared contributions without strong statistical support. PC-based representations distributed importance across multiple axes without a dominant structural dimension. Collectively, these results indicate that even where prediction is strong (e.g., complexity), stability emerges from combinations of structural properties rather than single dominant descriptors.

3.4.4 Differential Effects Across Stability Components

- Heatmaps
- Cross-component comparisons

4 Discussion

4.1 Why consensus on "the best metric" has remained elusive

For more than four decades, the structure–stability debate has oscillated between competing structural descriptors, from connectance and complexity to modularity, omnivory, and trophic redundancy. Early theoretical work, most notably by May, suggested that increasing complexity should destabilise large systems, placing global descriptors such as connectance and interaction density at the centre of the debate. Subsequent empirical and theoretical studies, however, demonstrated stabilising roles for omnivory, weak links, and compartmentalisation (e.g. McCann; Dunne), shifting attention toward pathway-level and geometric properties. Our results suggest that this apparent inconsistency does not arise because some metrics are "wrong," but because they capture different dimensions of a fundamentally multi-layered architecture. Food-web structure is not a single axis ranging from simple to complex; it is a high-dimensional space in which node composition, pathway routing, geometric constraint, and emergent organisation vary semi-independently. Metrics that have historically been treated as competitors—such as richness, connectance, omnivory, and redundancy—are in fact anchors of distinct structural dimensions. This explains why no single metric consistently predicts "stability" across studies: stability itself is not singular. When studies measure persistence (e.g. secondary extinctions), node-level balance and redundancy dominate. When they measure local stability or damping, path-level coupling and basal channel dominance matter more. When they measure recovery or informational organisation, global heterogeneity and role differentiation become decisive. The search for a universal structural predictor of stability has therefore been misguided—not because structure is unimportant, but because both structure and stability are multidimensional.

4.2 Reconciling contrasting structure–stability relationships

A persistent challenge in the literature has been that similar metrics sometimes show opposite associations with stability. For example, omnivory has been reported as stabilising in some contexts and destabilising in others. Within our framework, this variability is expected. Path-level coupling (e.g. omnivory motifs) primarily relates to resistance—the containment or diffusion of perturbations—rather than persistence per se. A structure that dampens oscillations may not prevent extinction cascades, and vice versa. Similarly, our finding that richness contributes substantially to structural variance but fails to predict any stability component underscores an important decoupling: network size is not synonymous with robustness. This result aligns with empirical syntheses showing that species richness alone poorly predicts secondary extinction resistance unless accompanied by trophic balance or redundancy. In our analyses, the predator–prey ratio and

herbivory—metrics reflecting trophic shape and basal channel dominance—were far stronger determinants of persistence and damping than richness itself. This helps reconcile why biodiversity–stability studies sometimes find positive, neutral, or context-dependent relationships. Richness operates primarily as a geometric scaling variable. Its effect on stability depends on how additional species alter the configuration of energy channels—whether they add redundancy, elongate chains, skew trophic balance, or increase coupling. Without specifying which structural dimension richness modifies, its stability effect is indeterminate.

4.3 Implications for biodiversity–stability theory

Our results contribute to a long-standing tension between the "complexity begets instability" argument and more recent views that diversity enhances resilience. The resolution lies in recognising that different dimensions of complexity influence different stability components. Redundancy and trophic similarity (low MaxSim heterogeneity) enhance persistence by providing functional compensation following species loss. Pathway multiplicity and basal channel dominance regulate resistance by shaping how perturbations propagate. Global organisational heterogeneity (captured by SVD complexity and spectral properties) influences return dynamics by constraining system-wide oscillations and reassembly patterns. In this sense, complexity is neither inherently stabilising nor destabilising. Instead, distinct forms of complexity map onto distinct dynamical outcomes. Connectance, motif structure, redundancy, and informational heterogeneity are not interchangeable proxies for "complexity," but structurally and mechanistically distinct attributes. By explicitly embedding food-web metrics within an energy-flow hierarchy, our framework reframes biodiversity–stability theory. Stability emerges not from species number alone, but from the configuration of energy channels and the distribution of functional roles across those channels. A bottom-heavy web with balanced predator–prey ratios and differentiated trophic niches may achieve both persistence and damping, even at moderate richness. Conversely, a species-rich but top-heavy or weakly differentiated web may remain fragile.

4.4 Structural modularity and the minimum sufficient set

A key practical outcome of our analysis is the identification of six structural pillars that capture nearly 85% of the variation in network architecture. Despite starting with 31 commonly used metrics, much of their variance collapses into a smaller number of independent axes. Hierarchical clustering further revealed robust modules corresponding to trophic balance, pathway structure, geometric scale, and redundancy. This dimensional reduction has two important implications. First, it demonstrates that many commonly reported metrics are statistically redundant. Reporting both connectance and L/S, or multiple correlated distance measures, adds little new information. Second, it provides a principled route toward a minimum sufficient

descriptor set tailored to ecological questions. Rather than selecting metrics by convention, researchers can select representatives of the relevant structural scale.

4.5 Guidance for metric choice based on ecological question

If the question concerns extinction cascades or tolerance to species loss (persistence): prioritise node-level balance and redundancy metrics (e.g. predator–prey ratio, herbivory, similarity). If the question concerns perturbation propagation or damping (resistance): focus on path-level and basal-channel metrics (e.g. pathway multiplicity, omnivory, mean distance). If the question concerns recovery dynamics or organisational reassembly (return): emphasise global heterogeneity and spectral measures (e.g. SVD complexity, spectral radius). If the question concerns comparative architecture across systems: use geometric anchors (e.g. richness) but interpret them as scale descriptors rather than stability predictors. This scale-explicit approach shifts the emphasis from asking "Which metric best predicts stability?" to "Which structural dimension corresponds to the stability mechanism of interest?"

4.6 Limitations and future directions

While our framework clarifies structural dimensionality, it remains grounded in static network topology. Real ecosystems exhibit interaction strengths, temporal variability, adaptive rewiring, and environmental forcing. Incorporating weighted networks and dynamic simulations would strengthen the mechanistic link between structural axes and realised dynamics. Additionally, the separation of persistence, resistance, and return is analytically useful but biologically intertwined. For example, structures that enhance resistance may indirectly support persistence by preventing large cascades. Future work could explore how these components interact across environmental gradients and disturbance regimes. Finally, extending this framework beyond trophic networks to mutualistic, host–parasite, or multilayer networks would test its generality. The energy-flow hierarchy proposed here may represent a broader organising principle of ecological interaction systems.

5 Conclusion

Food-web stability is not governed by a single structural property, but by multiple, hierarchically organised dimensions of network architecture. Node composition, pathway routing, geometric constraint, and emergent organisation each represent distinct levers through which ecosystems maintain persistence, resist perturbation, and reorganise following disturbance.

By explicitly mapping structural scale to stability mechanism, we reconcile decades of seemingly conflicting

findings in the structure–stability literature. The key insight is not that one metric is superior, but that different metrics answer different ecological questions. Recognising and embracing this multidimensionality provides a coherent foundation for future biodiversity–stability research and a practical framework for selecting network descriptors based on mechanism rather than tradition.

Table 4: Network properties used for analysis.

Dimension	Key Metrics	Expected Effect on Stability	Supporting Literature
Complexity & Redundancy	Connectance, MaxSim, Links	**Positive:** High redundancy allows for 'functional compensation' if one species is lost.	Dunne et al. (2002); McCann (2000)
Compartmentalization	Clust, Modularity,	**Positive:** Limits the spread of perturbations; local collapses don't become global.	Stouffer & Bascompte (2011)
Feedback & Coupling	Omnivory (S2), Loop, ChLen	**Variable:** Omnivory can stabilize by diffusing energy, but long chains can amplify oscillations.	McCann (2000); Neutel et al. (2002)
Hierarchy & Shape	Prey:Predator, Basal, Top	**Critical:** 'Bottom-heavy' systems are generally more stable; inverted pyramids are fragile.	
Information Heterogeneity	SVD Complexity, LinkSD	**Positive:** Diverse interaction strengths prevent 'resonant' instabilities.	Ulanowicz (2001)

To better link network structure to ecosystem function, we grouped commonly used food web metrics into four functional categories reflecting their ecological interpretation. Node-level metrics (e.g., basal, top, intermediate, herbivory, cannibal, trophic level, centrality, MaxSim) capture the roles and properties of individual species. Path-level metrics (*e.g.,* food chain lengths, motifs, omnivory, loops, prey-to-predator ratios) describe how energy and interactions flow through the network. Geometry metrics (e.g., connectance,

link density, clustering, variation in links, diameter, intervality, spectral radius, SVD complexity) quantify the overall network arrangement and topological organization. Finally, behavioural/system-level metrics (e.g., robustness, spectral radius, SVD complexity) capture emergent properties such as resilience, redundancy, and dynamic stability. This framework provides a transparent, ecologically interpretable mapping from species-level roles and interaction paths to system-level behaviour, facilitates identification of a Minimum Sufficient Set of descriptors, and links network structure to long-standing questions in the stability–complexity debate.

Table 5: Classification of food web metrics into functional categories. Node-level metrics capture species roles; path-level metrics capture energy or interaction flows; geometry metrics describe network structure; and behaviour/system-level metrics capture emergent stability and resilience.

Category	Metrics	Rationale
Node-level	basal, top, intermediate, herbivory, cannibal, TL, centrality, MaxSim	Metrics describing individual species and their roles in the network, including trophic position (TL), feeding type (basal, herbivory, cannibal), influence (centrality), and functional redundancy (MaxSim).
Path level	ChLen, ChSD, ChNum, path, S1, S2, S4, S5, omnivory, loops, predpreyRatio, distance	Metrics describing energy flow or interaction paths through the network, including food chain lengths, motifs, omnivory, loops, and prey-predator ratios. Capture how energy and interactions move across nodes.

Category	Metrics	Rationale
Geometry/topology	connectance, l_S, links, richness, Clust, GenSD, VulSD, LinkSD, diameter, intervals	Metrics describing **overall network structure**, including density (connectance, links, L/S), size (richness), local clustering, link asymmetry, network distances (diameter), and niche structure (intervals). Capture the 'shape' of the network.
Behaviour/system-level	, complexity, robustness	Metrics describing emergent properties of the network, including system-wide stability, resilience, and structural complexity. Capture how the network responds to perturbations or organizes itself at a global level.

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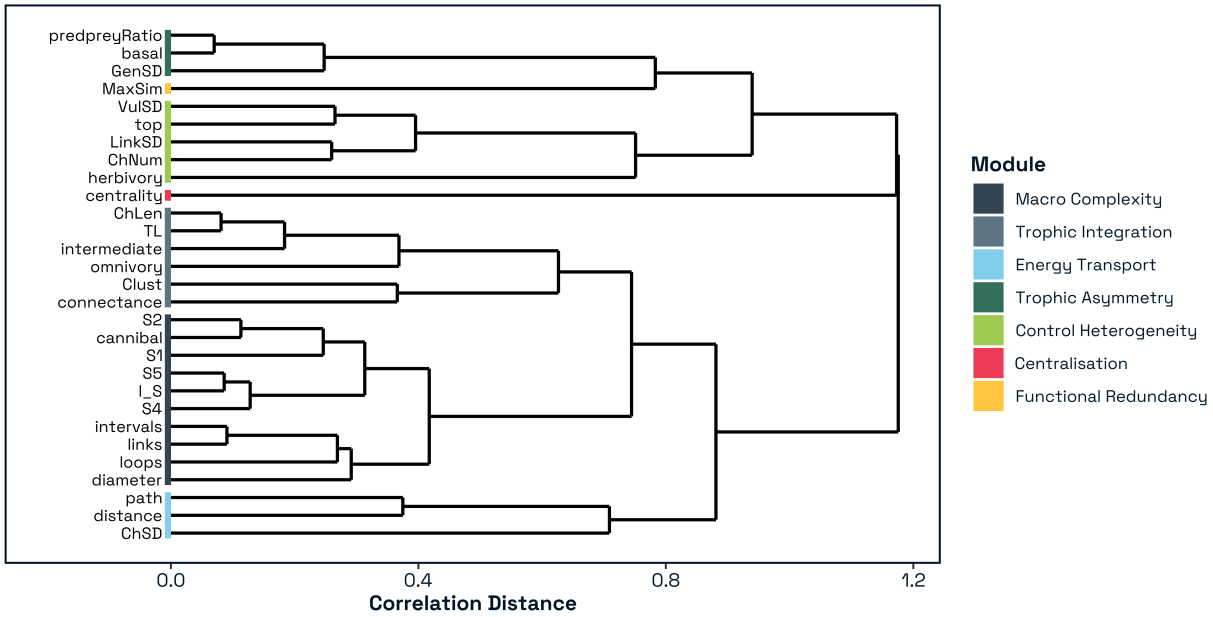

Figure 1: Blah blah blah clusters

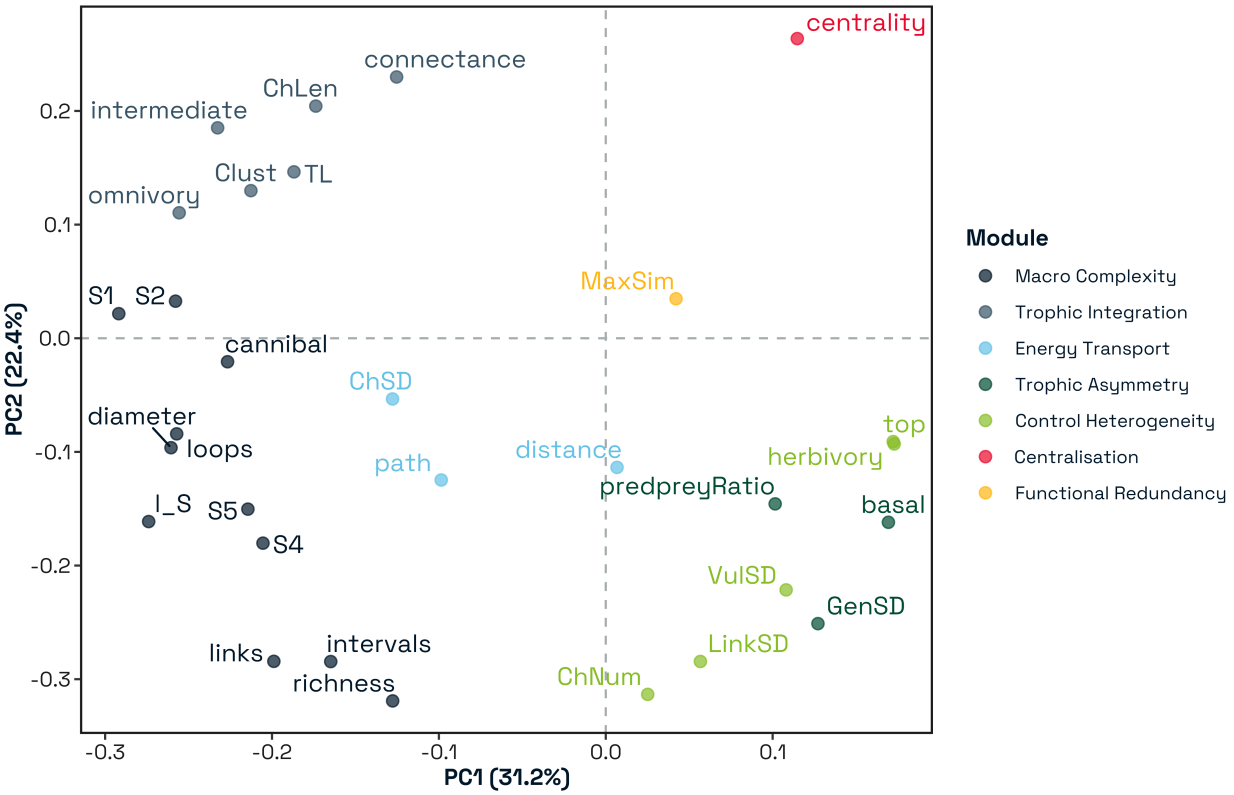

Figure 2: Blah blah blah - colours represent clusters. Network metric loadings with PC1 and PC2

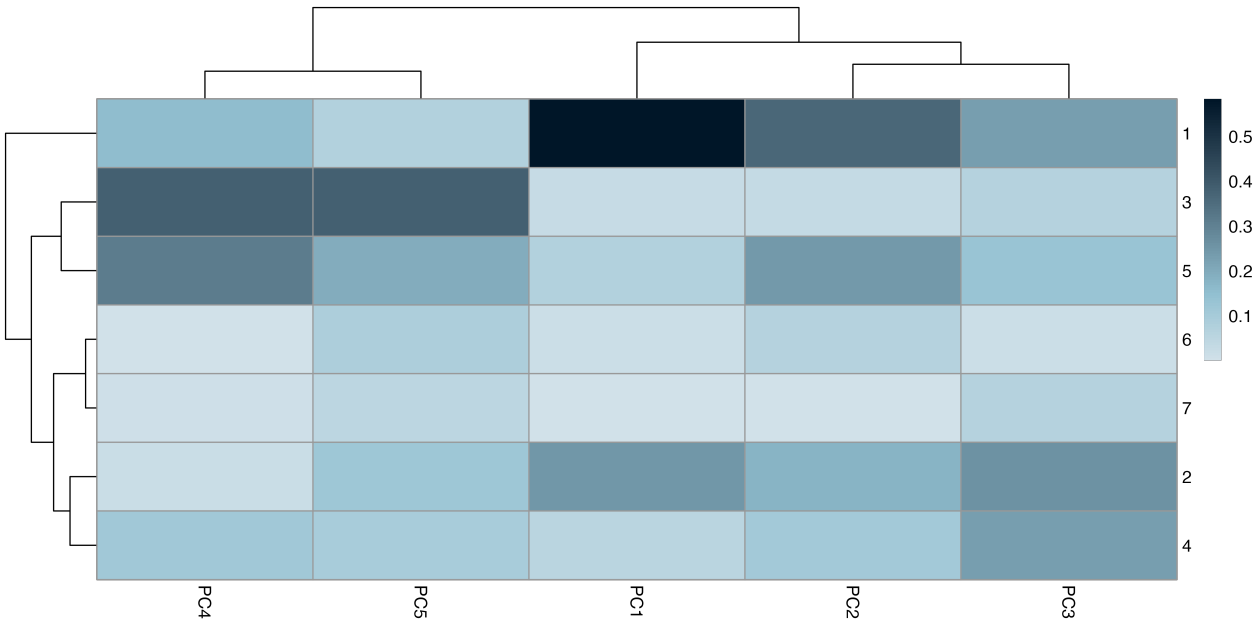

Figure 3: Blah blah blah - colour intensity = greater alignment

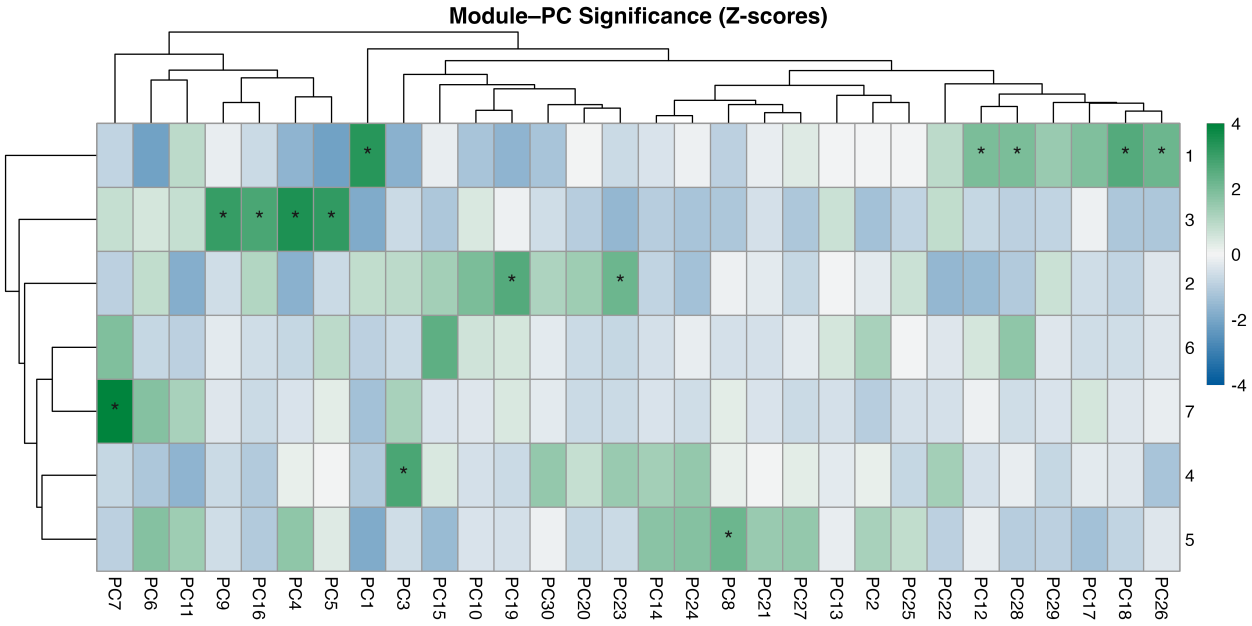

Figure 4: Heatmap showing module–principal component alignment quantified by permutation-derived Z-scores. Colour intensity reflects the magnitude and direction of alignment, and asterisks denote cells that meet joint significance criteria (permutation $P < 0.05$ and $|Z| > 1.96$). Rows correspond to structural metric modules and columns to principal components. Clustering highlights patterns of shared alignment across modules and multivariate structural gradients.

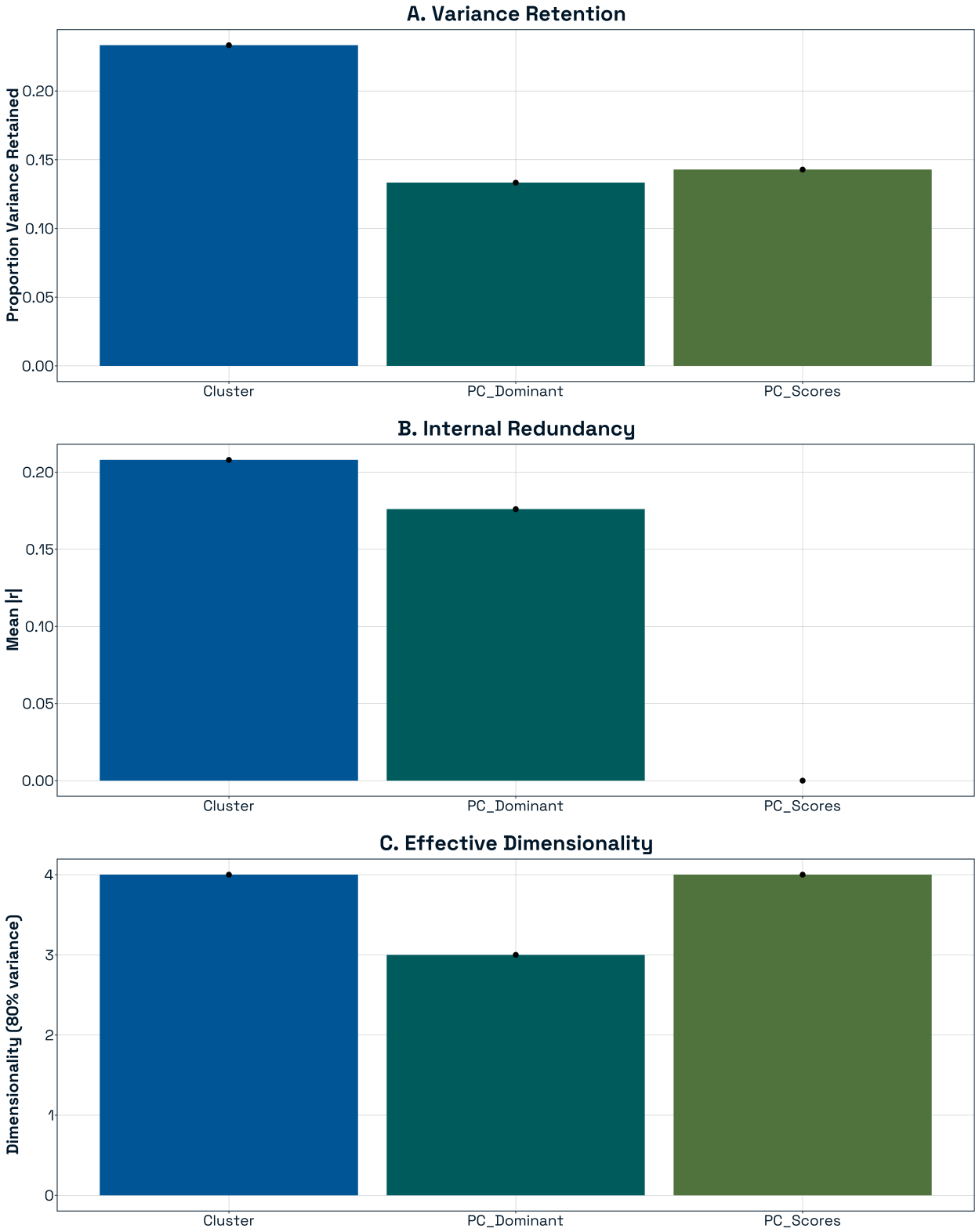

Figure 5: Figure S1. Characterisation of alternative dimensional-reduction approaches for network structural metrics. (A) Proportion of total variance in the full metric space retained by each representation: cluster medoids, PC-dominant metrics, and retained PC scores (80% cumulative variance). (B) Mean absolute pairwise correlation (|r|) among predictors within each representation, quantifying internal redundancy. (C) Effective dimensionality, defined as the number of axes required to explain 80% of variance within each reduced predictor set. Together, these diagnostics illustrate differences in information retention, redundancy, and structural alignment among dimensional-reduction strategies.

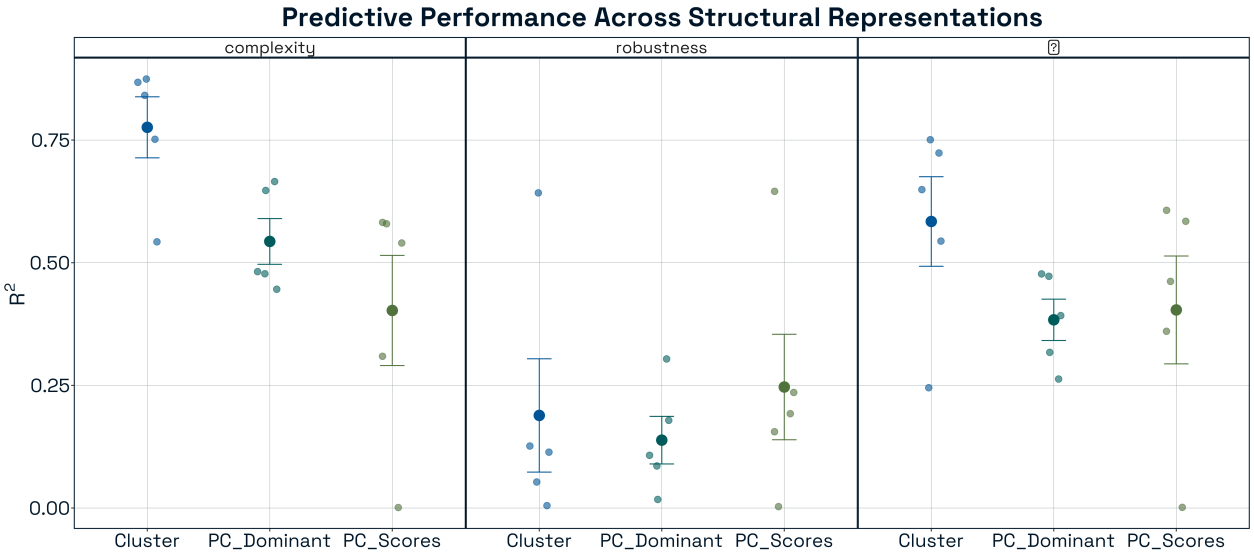

Figure 6: Figure S2. Cross-validated predictive performance (R^2) of random forest models across structural representations. Points represent individual cross-validation folds; large points indicate mean ± SE. Predictive performance varied among stability components, with cluster medoids substantially outperforming PC-based representations for complexity but not for robustness or .